Raspberry Pi Pico

P1
rpi pico 1 W
2648-SC0918CT-ND
smd

+3.3V
R1 220
3V1 green
GND

R10 4.7k
GND

+BATT
+3.3V
5V
J1

NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2

D1 D_Schottky
5V
+3.3V

R2 220
D13 blue
GND

bt1
bt2
bt3
bt4

sda
scl

right_2
right_1
right_en

left_2
left_en
left_1

J3 joystick
J4 speed knob

3V3_EN
RUN
VBUS
VSYS
3V3_OUT
ADC_VREF
GP0
GP1
GP2
GP3
GP4
GP5
GP6
GP7
GP8
GP9
GP10
GP11
GP12
GP13
GP14
GP15
GP16
GP17
GP18
GP19
GP20
GP21
GP22
GP26
GP27
GP28
AGND
GND
GND_1
GND_2
GND_3
GND_4
GND_5
GND_6

* these pins are 5V tolerant inputs.
The input is divided down from 5v to a safe level.
If you want to use this pin as a 3.3v input or as an output,
solder the NetTie pads on the back of the board.

left motor

U1
VNH7040AYTR

HSL1
Heatsink

J100 0197054001
J101 0197054001
optionally replace J7 to be weelye compatible

J7
1017491

+BATT
C1 0.1uF
GND

ML+
ML-

NC
Drain_LSD_A
GND_A
Drain_LSD_B
VCC
Source_HSD_A
Source_HSD_B
SEL0
IN_A
PWM
NC
Multisense
SEL1
IN_B
Multisense_En

left_2
left_en
left_sense

right motor

U2
VNH7040AYTR

MR+
MR-

C2 0.1uF

right_1
right_en
right_sense

J8
1017491

optionally replace J8 to make the PCB weelye compatible
weelye style J104
weelye style J105
J106 0197054001
J107 0197054001

HSR1
Heatsink

ADC (for measuring battery voltage and motor current)

U4
TLA2024IRUGR

+3.3V
NT1
NetTie_3
TP0
TestPoint
+BATT
R13 100k
R14 8.2k
GND

ADDR
AIN0
AIN1
AIN2
AIN3
NC
VDD
SCL
SDA
right_ipropi
left_ipropi
scl
sda

C4 0.1uF
pullup resistors

left_sense
right_sense
R17 220
R18 220
R11 442
R12 442
D6 3V zener

R15 3.3k
R16 3.3k

qwiic I2C plug
J2 Conn_01x04

weelye compatible header
J11 Conn_01x10_Pin
switch
bat-
D2 D_Schottky

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1 ASJ-200-B-HT
B2 ASJ-200-B-HT
B3 ASJ-200-B-HT
B4 ASJ-200-B-HT

battery power circuit
J102 0197054001
J103 0197054001
optionally replace J5 to be weelye compatible
J5 TB005-762-02BE
screw terminal
SW1
switch
R3 3.3k
R4 22.1k
R5 22.1k
18V zener
D4
D3
PSMN1R0-40YLDX
Q1 on/off switch
PSMN1R0-40YLDX
Q2 reverse polarity protection
C5 33uf
J6

alternative voltage regulator
Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.
U5
D78B05T, RBT05W24S05, pololu5593

voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=49A4BE669EF8F291
C6 0.1uF
U3
LMR54410DBVR
L1 10uH
C3 1uF
C7 0.1uF
R8 115k
R9 22.1k
C8 22uF

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1 M3
H2 M3
H3 M3
H4 M3

M1
M2
optional cooling fans
wire to J6
MF40200V1-1000U-A99 (5V, 3.5-6)
MF40201V1-1000U-A99 (12V, 4.5-13.8)
MF40202V1-1000U-A99 (24V, 8-27.6V)

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com
Sheet: /
File: gbg-pcb.kicad_sch
Title: GBG-PCB
Size: A3 | Date: | Rev: v2.4
KiCad E.D.A. 10.0.8 | Id: 1/1